Week 47 Gentle step landscapes

Facilitator guide | Student packets v5 | Prepared 3 October 2026, revised 23 October 2026 | Unpiloted

Mathematical overview

Bounded change on a row. Sites are consecutive integers; heights are nonnegative integers, and adjacent heights differ by at most 1. There is at least one fixed clue (c, a_c). A completion exists exactly when $|a_c - a_d| \le |c - d|$ for every pair of clues. Distance means the number of steps, not the number of sites visited.

When the pairwise condition holds, the simultaneous least and greatest completions are
$L(i) = \max(0, \max_c(a_c - |i - c|))$ and
$U(i) = \min_c(a_c + |i - c|)$.
Every legal completion lies between them, and both envelopes are themselves legal; thus they also minimize and maximize total height. Without the condition the formulas are not landscapes: for P4's first set, h(0)=0 and h(2)=3, they give L = (1,2,3,2,1) and U = (0,1,2,3,4), each breaking a clue. No fixed supply of blocks or extra height ceiling is assumed. With no clues there is no greatest completion.

Moves. Changing one free height by one level per move and keeping every intermediate row legal, the fewest moves between two landscapes with the same clues always equals the total change, the sum of |h(i) - g(i)|. Some orders are illegal, but a legal one always exists: among the sites above the target, a highest one can always come down a level, and symmetrically for sites below. K-1 P6 (5 moves) is an instance.

Discovery by band. K-1 builds legal landscapes, tests impossible jumps and finds a shortest sequence of one-level moves. Grades 2-3 links allowable heights to distance and optimizes a whole landscape. Grades 4-5 shades allowed heights, finds that sitewise choices need not fit together, and explains why pairwise-consistent clues can always be filled. Experiments support conjectures; the distance and envelope arguments establish them.

Prepare and launch

Give each child their chosen packet, a pencil, and 7 white and 3 black flat counters about 1.5 cm across. Black counters are the clues: they sit on the printed black discs, and they move off those discs when a problem moves or replaces a clue (K-1 P3 and Grades 2-3 and 4-5 P3, on the Problem 2 board; K-1 P7 on its own empty board). Flat counters keep zero visible and avoid a large cube supply. Main boards have 22 mm site spacing and 20 mm level spacing; the Problem 4 board uses 25 mm site spacing, and the Grades 4-5 Problem 6 boards 16 mm levels for pencil shading. The small rows beside the Grades 2-3 and 4-5 Problem 1-2 boards are for writing finished landscapes. Print at actual size. The drawing’s top edge is not an extra mathematical restriction.

Readiness. K-1 can compare small whole heights with adult reading. Older entries require counting steps, then adding/subtracting distances; formulas are adult shorthand. With the current eleven-child group (four at K-1, four at 2-3, three at 4-5), keep the three familiar adult-led tables and let readiness determine the page.

Launch, 4 minutes, after the five-minute run. Let children place markers, then follow the printed 1,2,2,1 towers to one marker per column and the boxed row, always left to right. Contrast the illegal 1,3,2 jump. Point to the fixed clues and visibly represent height zero. Ask, “Can you change a white marker and keep every neighboring jump small enough?” Do not demonstrate the envelope recipe yet.

Flexible hour. 0-5 running; 5-9 launch; 9-47 at the tables, constructing and comparing first, then optimization or partner puzzles; 47-55 share one construction and one impossibility reason; 55-60 reset. Completing the packet is not the goal.

Optional hints and stopping points

For a stuck construction, ask what one neighbor permits; for impossibility, ask how many steps separate the clues. For an extremum, cover all but one clue and ask what that clue alone forces. For K-1 moves, compare the three starting and target heights before trying to prove optimality. Stop with a convincing construction and a local jump explanation; do not require the envelope notation from children.

Key for the shared starting problems

Problem 1, all bands. Endpoints are 1 at sites 0 and 4. There are 18 landscapes; Grades 2-3 and 4-5 write four in the rows beside the board, for example (1,0,0,0,1), (1,1,1,1,1), (1,2,1,2,1), (1,2,3,2,1). All adjacent differences are at most 1. The middle can be 0 and can be 3. Its full possible range is 0-3, since it is two steps from either clue. Any legal different constructions answer K-1.

Problem 2, all bands. The star is site 2; clues are h(0)=0 and h(4)=2. Its possible heights are exactly 0, 1, 2. Witnesses, one for each height, are (0,0,0,1,2), (0,0,1,1,2), (0,1,2,2,2); two of the five record rows stay empty. The left clue bounds it above by 2; nonnegativity bounds it below by 0. K-1 asks only for the extrema.

Problem 3. Children use the Problem 2 board, moving the right black counter off its printed disc. K-1's new right clue is 4: the only landscape is (0,1,2,3,4). In Grades 2-3 and 4-5, 4 is the only right-clue height that forces every site. Larger heights are impossible. For heights r=0,1,2,3, there is slack: the least completion and the greatest completion differ, so at least two landscapes exist.

Problem 4, all bands, from top to bottom. First impossible: heights 0 and 3 are only two steps apart. Second possible: (3,2,1,0,0) is a completion. Third impossible: heights 0 and 3 are again only two steps apart. Each impossibility follows by adding two permitted jumps, whose total change cannot exceed 2.

Key for the seven site landscape

Problem 5, all bands. Clues are h(0)=1, h(4)=3, h(6)=1. The least landscape is **(1,0,1,2,3,2,1)**, total 10; the greatest is **(1,2,3,4,3,2,1)**, total 16. Each marker is bounded by its distance from each clue. Both displayed landscapes attain all the respective bounds at once.

For K-1, a third different answer is (1,1,1,2,3,2,1); many answers work. There are exactly 10 completions, independently enumerated, but no printed task asks children to list them all. Grades 2-3 asks for the smallest and largest total heights, 10 and 16. Grades 4-5 asks whether one landscape is lowest at every site at once, and one highest: yes, these two. The same two constructions solve both questions.

Key for the K-1 finish

Problem 6. The minimum is **5 moves**. A legal route is
(1,2,1,2,1) → (1,1,1,2,1) → (1,1,1,1,1) →
(1,1,0,1,1) → (1,0,0,1,1) → (1,0,0,0,1).
The three white heights must fall by 2, 1 and 2. One move changes just one height by one level, so fewer than 5 moves cannot suffice. Fixed end clues never move. Lowering the middle first gives (1,2,0,2,1), with two jumps of 2, so the order matters. Partners: one child moves a marker; the other checks both jumps beside it before the next move.

Problem 7. The page 6 board is empty, with sites 0-4 and levels 0-4. Clues h(0)=0 and h(4)=4 force (0,1,2,3,4), and their mirror forces (4,3,2,1,0); no other pair of clues on this board forces one landscape. Clues h(0)=h(4)=1 allow many answers. Check a child's invented pair by height difference ≤ site distance. An impossible puzzle is a valid investigation if it is identified and explained as such.

Key for the older finishes

Grades 2-3 Problem 6. On sites 0-6, h(0)=0 and h(6)=6 force (0,1,2,3,4,5,6). To gain 6 in six steps, every step must rise by 1. The pair h(0)=h(6)=0 allows the all-zero landscape and (0,1,0,0,0,0,0), so it allows more than one. On this board these are the only forcing pairs, with the mirror h(0)=6, h(6)=0. A clue away from an end leaves the end site at least two choices, so both clues must sit at sites 0 and 6, six levels apart. For example, h(1)=0, h(5)=4 fixes sites 1-5 but still allows 6 landscapes.

Grades 2-3 Problem 7. A clue four steps from height 1 can have any whole height **0,1,2,3,4,5**. In general, d steps from a clue of height a, the possible heights are max(0,a-d) through a+d, inclusive, when there are no other clues. Join two allowed clue heights by changing one level until the second height is reached, using flat steps for the remaining distance. Other clues can narrow this range.

Grades 4-5 Problem 6. First board, one clue h(3)=2: sites 0 and 6 allow **0-5**, sites 1 and 5 allow 0-4, sites 2 and 4 allow 1-3 (from max(0, 2-d) to 2+d at distance d). Second board, h(0)=0 and h(5)=3: sites 1-4 allow **0-1, 0-2, 1-3, 2-4** and site 6 allows **2-4**, the overlap of each clue's range. Picks need not fit: 0 at site 2 and 3 at site 3 are each allowed but never together, as are 4 at site 1 and 1 at site 2 on the first board. "Possible at a site" is a separate existence claim. Yet the lowest picks together, and the highest together, do make landscapes: they are Problem 5's envelopes.

Grades 4-5 Problem 7. Yes, under the printed nonnegative whole-height rule. Use the lower envelope L from page 1. Each clue contributes a downward tent $a_c - |i-c|$; take the largest of these and zero at every site. The maximum of these 1-Lipschitz functions is 1-Lipschitz. At a clue d, its own tent equals a_d; pairwise consistency makes every other tent no larger. Thus L has the required clue values and is nonnegative and integral. This is a complete construction, not only a feasibility test. The page 6 board is for trying clue sets.

For completeness, the upper envelope U is also 1-Lipschitz: if every upward tent $f_c(i) = a_c + |i-c|$ changes by at most one between adjacent sites, their finite minimum does too. Pairwise consistency makes U hit each clue. Every completion h must satisfy $a_c-|i-c| \le h(i) \le a_c+|i-c|$, proving the extremal claims.

Sources and verification

E. J. McShane, *Extension of range of functions*, Bulletin of the AMS 40 (1934), 837-842, original paper DOI. The original PDF was inaccessible during preparation. The extension formula was checked in N. Gigli and E. Pasqualetto, *Equivalence of two different notions of tangent bundle on rectifiable metric measure spaces*, §1.2, equation (1.16), manuscript p. 7: author manuscript. The finite integer proof and keyed examples above were independently checked.

N. Rozhkovskaya, *Math Circles for Elementary School Students*, "Introduction: Berkeley 2009," describes individual adult support and active parent involvement. The three-table plan here is a local proposal. This guide is unpiloted; digital enumeration and PDF dimension checks are not physical or classroom tests.

Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

Use and return choices

First routes in all three bands: P1-2; add the forced-ramp question P3 when children can check every neighboring jump. On return, K-1 can build the seven-site landscapes (P5) or change one marker per move with a partner checking each jump (P6) and invent clues on the page 6 board (P7). Grades 2-3 can optimize totals (P5), then make partner clues or a distance rule (P6-7). Grades 4-5 can build pointwise envelopes (P5), shade allowed heights around one clue and around two (P6), and justify completion from all pairwise bounds (P7). Keep zero visible and retain child-chosen constructions before presenting a general recipe.